

Education

Bachelor of Science in Computer and Information Science; Bachelor of Arts in Chinese

Specialization: *Information & Computational Assurance*

The Ohio State University | Columbus, OH

August 2022 – December 2026

- GPA: 3.86 (Dean's List all terms) - student member of ACM (htrowbridge@acm.org)
- Relevant Coursework: Data Structures and Algorithms, Information Security, Network Security, Computer Networking

Work Experience

IT Intern – Enterprise Cybersecurity Policy

Abbott Laboratories

Chicago, IL

May 2026 – August 2026

- Spearheaded an end-to-end, proof-of-concept internal cyber policy AI assistant, delivering 40+ pages of engineering documentation and designing a 50-question audit suite that validated 86% model accuracy with 9 SMEs
- Accelerated global control-objective modernization by designing a proposal to consolidate and clean-up legacy controls, reducing redundancies, easing adoption, and maximizing auditability
- Executed a preliminary gap analysis mapping internal cybersecurity controls against NIST SP 800-218 secure software development lifecycle (SSDL) standards
- Evaluated 3 cloud-integrated AI transcription models for a call-center dashboard by engineering 4 novel metrics to optimize data capture and triage; proposed adoption of a more performant alternative projected to reduce operating costs by 75%

Undergraduate Research Assistant (Industrial Cybersecurity Lab)

Center for Design and Manufacturing Excellence

Columbus, OH

May 2025 – Present

- Co-architected a full-stack compliance platform using Flask, Flask-Security-Too, and PostgreSQL, implementing strict authentication and RBAC to help defense manufacturers operationalize NIST SP 800-171r3 (CMMC 2.0 Level 2) compliance
- Developed an asynchronous computer-vision platform using FastAPI/Uvicorn and PostgreSQL in Podman, streamlining image ingestion for automated shoe-wear evaluation under Agile/Scrum
- Benchmarked Hyperledger Fabric state-pruning algorithms, quantifying throughput, latency, and storage reduction at scale

Software and Data Engineer Intern

GovRat

Washington, D.C. (Remote)

September 2025 – May 2026

- Engineered Python scraping pipelines to ingest and normalize federal contract datasets, improving downstream model quality

Classroom and Digital Union Support Technician

The Ohio State University Office of Technology and Digital Innovation

Columbus, OH

March 2025 – September 2025

- Resolved mission-critical IT/AV failures in classrooms, ensuring uninterrupted instruction for hundreds of students weekly

Accounting Intern

Jeffrey W. Krol & Associates, Ltd.

Chicago, IL

June 2023 – August 2023; May 2024 – July 2024

- Built digital archival system and Excel dashboards, reducing reporting turnaround time and improving investment tracking

Technical Projects and Research

U.S. Cyber Command Research Competition: Cyber RECon 2026's Winning Paper

Semantic Shields: Automating Critical Infrastructure Defense via NLP-Driven Ransomware Profiling

Fort Meade, MD

April 2026

- First-authored, published research paper proposing NLP-driven ransomware profiling to automate threat detection across U.S. critical infrastructure sectors
- Won all 3 competition categories at U.S. Cyber Command's Cyber RECon 2026: Overall Best Research, Best Paper (Analyst Track), and Best Presentation

Debian Package Mirrors

June 2025 – February 2026

- Deployed Debian mirrors on FreeBSD (complete) and Debian with NGINX, provided reliable package access for users
- Automated synchronization with official repositories, ensuring updated packages with minimal downtime

Certifications and Qualifications

- Centrix™ Power User | Armis | Centrix™ Power User Learning Path (7.5 CPE)
- Security+ (in-progress) | CompTIA

Leadership

- Vice President, [Cybersecurity Club](#) | Co-organized BuckeyeCTF for 200+ national competitors; developed weekly meeting content
- Vice President, [Open Source Club](#) | Directed a cross-org partnership with Ohio LinuxFest; led technical workshops promoting FOSS
- Member, [History Club](#) | Lectured on Three Kingdoms–era Chinese history; regular attendee

Skills

- Security & Compliance:** NIST SP 800-171r3, CMMC 2.0 (Level 2), NIST SP 800-218 (SSDF), Zero Trust Architecture, Threat Intelligence, Access Control Lists (ACLs), Principle of Least Privilege, AI Risk & Model Evaluation, Gap Analysis
- Systems & Network Security:** Linux (Debian, RHEL, Fedora), FreeBSD, Firewalls (UFW, firewalld), Network Security, Tailscale (Mesh VPN), Cloudflare, NGINX, SSH, RDP
- DevSecOps & Infrastructure:** Docker, Podman, Container Security, Git, Process Automation, Web Server Hardening
- Languages & Data Engineering:** Python (FastAPI, Flask, SQLAlchemy, Pandas, Plotly), SQL (PostgreSQL, SQLite), Bash/POSIX Shell, C, Java, Ruby on Rails, REST APIs, Web Scraping
- Languages (Spoken):** English (Native), Mandarin Chinese (Professional Working Proficiency)